DeepSeek-V2

Efficient MoE Language Model with Multi-Head Latent Attention

236B Total Parameters · 21B Activated · 128K Context

arxiv.org/abs/2405.04434

2024

PROBLEM FRAMING

The LLM Scaling Dilemma: Performance vs. Training Cost and Inference Speed

THE PROMISE

Larger models yield better intelligence

↑ **Scale**

More parameters = More capability

→

THE PROBLEM

Prohibitive costs and slow inference

↑ **Cost**

KV cache bottleneck

→

THE SOLUTION

Efficient architectures MoE + Attention Innovation

✓

DeepSeek-V2: 236B total / 21B active

KEY INSIGHT

Standard MHA (Multi-Head Attention) stores full K/V per head per token — creating a memory bottleneck that grows linearly with context length. This KV cache problem makes long-context inference prohibitively expensive.

DeepSeek-V2 Architecture: MLA + DeepSeekMoE in a Transformer Block

Input Token →

MLA

Multi-Head Latent Attention

- Low-rank KV compression
- Decoupled RoPE
- 93.3% cache reduction
- Better than MHA performance

→

DeepSeekMoE

Mixture-of-Experts FFN

- Shared experts (always active)
- Routed experts (top-K selection)
- Fine-grained specialization
- Device-limited routing

→ **Output**

Multi-Head Latent Attention (MLA)

Instead of caching full K/V vectors per head, MLA compresses them into a compa latent vector c_t^KV. Full keys/values are reconstructed on-the-fly via learned projection matrices — achieving better performance than standard MHA while dramatically reducing the KV cache size.

DeepSeekMoE FFN

Fine-grained expert segmentation increases specialization potential. Shared experts capture common knowledge; routed experts are sparsely activated via top-K routing. Device-limited routing reduces cross-node communication overhead.

Multi-Head Latent Attention Compresses KV Cache by 93.3%

93.3% KV Cache Reduction

MHA
Multi-Head Attention

Full K/V per head

High memory

Best performance baseline

GQA
Grouped-Query Attention

Shared K, per-head V

Medium memory

Performance tradeoff

MQA
Multi-Query Attention

Shared K and V

Low memory

Performance degradation

MLA ★
Multi-Head Latent Attention

Low-rank KV compression

Minimal cache + Better perf

Decoupled RoPE

HOW MLA WORKS

1. Compress: Instead of caching full K/V vectors, compress into a compact latent vector c_t^{KV} ✓
2. Reconstruct: Full keys and values are recovered on-the-fly via learned projection matrices
3. Enhance: Decoupled RoPE-based positional key handles positional information separately

Better than MHA
with 93.3% less cache

MOE ARCHITECTURE

DeepSeekMoE: Fine-Grained Experts with Shared Expert Isolation

Shared Experts (N_s)

Always activated — capture common knowledge

S1 S2 S3 S4

Common knowledge across all tokens

Routed Experts (N_r)

Sparsely activated via top-K_r routing

Only top-K selected per token (fine-grained specialization)

Top-K Routing

Each token selects top-K experts via learned router — sparse activation reduces compute

Device-Limited Routing

Each token → experts on ≤M devices — reduces cross-node communication overhead

Auxiliary Losses

Load-balancing across experts and devices prevents expert collapse or underutilization

Token Dropping (Training)

Prevents expert overflow during training without affecting inference efficiency

RESULT

236B total parameters with only 21B activated per token → 42.5% training cost reduction vs. DeepSeek 67B dense model

21B

of 236B activated

EFFICIENCY GAINS

42.5% Training Cost Reduction and 5.76× Generation Throughput vs. De

DeepSeek 67B (Dense)	→	DeepSeek-V2 (MoE)
Training Cost **100%**	↓	Training Cost **57.5%** −42.5% reduction
KV Cache **High**	↓	KV Cache **Minimal** −93.3% reduction
Max Generation Throughput **1×**	↑	Max Generation Throughput **5.76×** higher

WHY IT MATTERS

Same performance as 67B dense m

93.3% KV cache reduction enables:

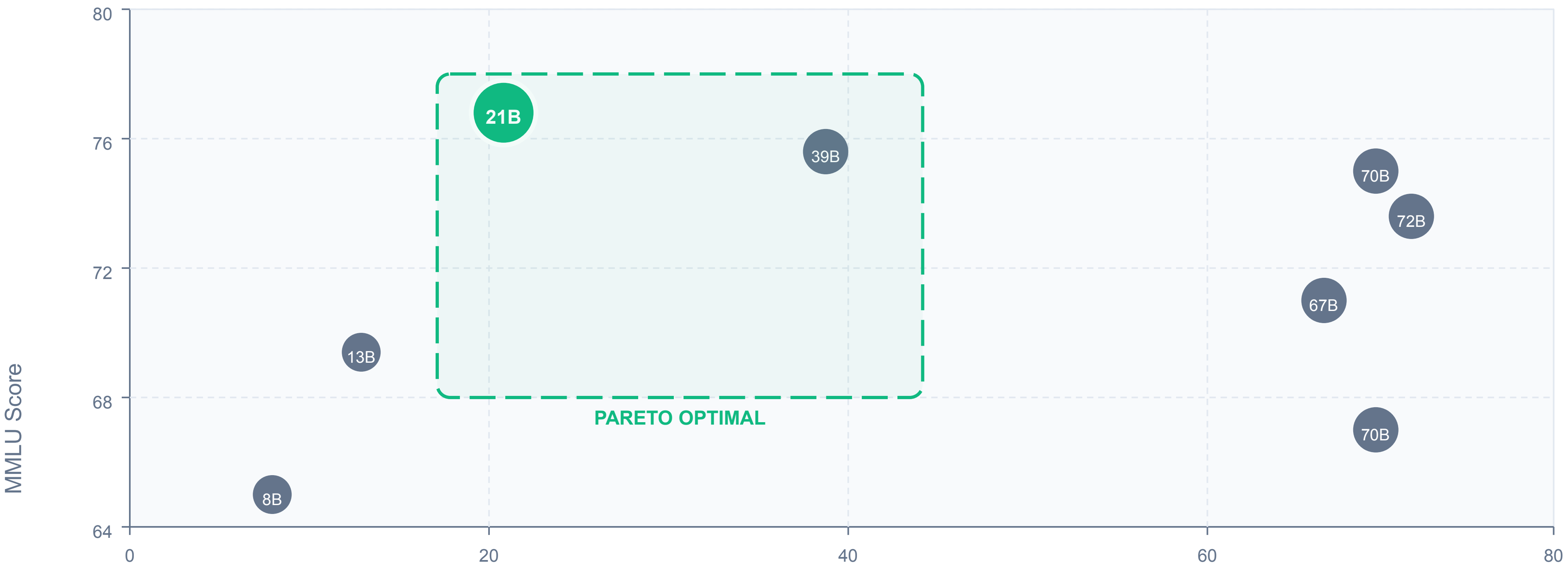
- Longer context windows
- Lower memory footprint
- Cheaper inference

5.76× throughput means:

- More requests per second
- Lower latency per token
- Better user experience

BENCHMARK

DeepSeek-V2 Tops MMLU Among Open-Source Models at Only 21B Activ



DeepSeek-V2
21B activated, 78.5 MMLU

LLaMA 3 70B
70B activated, 77.0 MMLU

Mixtral 8×22B
39B activated, 77.6 MMLU

Qwen1.5 72B
72B activated, 75.6 MMLU

DeepSeek 67B
67B activated, 73.0 MMLU

LLaMA 2 70B
70B activated, 69.0 MMLU

Mixtral 8×7B
13B activated, 70.6 MMLU

KEY INSIGHT

DeepSeek-V2 achieves the highest MMLU score while using fewer activated parameters

ALIGNMENT

DeepSeek-V2 Chat (RL) Rivals Closed-Source Models on Chinese Benchm

POST-TRAINING PIPELINE

Pretraining 8.1T tokens → **SFT** 1.5M sessions → **GRPO RL** Group Relative → **DeepSeek-V2 Chat** Production Ready

EVALUATION RESULTS

AlpacaEval 2.0

(length-controlled win rate)

38.9% win rate

MT-Bench

(overall score)

8.97 / 10

AlignBench

(Chinese, overall)

7.91 / 10

★

Strongest Open-Source Chat Model in Chinese

DeepSeek-V2 Chat (RL) outperforms all open-source models and most closed-source models on AlignBench, demonstrating exceptional multilingual alignment quality for both English and Chinese conversations.

GRPO (Group Relative Policy Optimization)

Reinforcement learning approach that uses group-relative baselines for stable training. Combines SFT's 1.5M conversational sessions with GRPO to achieve strong alignment across multiple evaluation benchmarks.

Key Takeaways: Architectural Innovation Unlocks Efficient Scaling

1. **MLA Superiority**

   Multi-Head Latent Attention achieves better performance than MHA while reducing KV cache by 93.3% — a strictly better alternative.

2. **DeepSeekMoE Efficiency**

   Fine-grained experts with shared expert isolation enables training 236B parameters at 42.5% lower cost than 67B dense model.

3. **Pareto-Optimal Performance**

   Only 21B activated parameters needed to match or exceed 70B-class dense models (LLaMA 3 70B, Qwen1.5 72B) on MMLU.

4. **5.76× Throughput Gain**

   Generation throughput 5.76× higher than DeepSeek 67B makes DeepSeek-V2 practical for large-scale deployment.

5. **Strong Multilingual Alignment**

   DeepSeek-V2 Chat (RL) is strongest open-source chat model in Chinese, competitive with closed-source systems in English evaluation.